

Bank

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.4.4      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.0
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tidymodels)
```

```
## -- Attaching packages ----- tidymodels 1.1.1 --
## v broom      1.0.5      v rsample    1.2.0
## v dials      1.2.0      v tune       1.1.2
## v infer      1.0.5      v workflows  1.1.3
## v modeldata  1.2.0      v workflowsets 1.0.1
## v parsnip    1.1.1      v yardstick  1.2.0
## v recipes    1.0.9
## -- Conflicts ----- tidymodels_conflicts() --
## x scales::discard() masks purrr::discard()
## x dplyr::filter()   masks stats::filter()
## x recipes::fixed() masks stringr::fixed()
## x dplyr::lag()      masks stats::lag()
## x yardstick::spec() masks readr::spec()
## x recipes::step()   masks stats::step()
## * Learn how to get started at https://www.tidymodels.org/start/
```

```
library(ggforce)
```

```
library(mctest)
```

```
library(olsrr)
```

```
##
## Attaching package: 'olsrr'
##
## The following object is masked from 'package:datasets':
##
##     rivers
```

```
library(jtools)
```

```
##  
## Attaching package: 'jtools'  
##  
## The following object is masked from 'package:yardstick':  
##  
##   get_weights
```

```
library(ggcorrplot)  
library(yardstick)  
library(car)
```

```
## Loading required package: carData  
##  
## Attaching package: 'car'  
##  
## The following object is masked from 'package:dplyr':  
##  
##   recode  
##  
## The following object is masked from 'package:purrr':  
##  
##   some
```

```
library(moments)  
library(GGally)
```

```
## Registered S3 method overwritten by 'GGally':  
##   method from  
##   +.gg   ggplot2
```

```
library(psych)
```

```
##  
## Attaching package: 'psych'  
##  
## The following object is masked from 'package:car':  
##  
##   logit  
##  
## The following objects are masked from 'package:scales':  
##  
##   alpha, rescale  
##  
## The following objects are masked from 'package:ggplot2':  
##  
##   %+%, alpha
```

```
library(fastDummies)
```

```
## Thank you for using fastDummies!  
## To acknowledge our work, please cite the package:  
## Kaplan, J. & Schlegel, B. (2023). fastDummies: Fast Creation of Dummy (Binary) Columns and Rows from
```

```
library(lmtest)
```

```
## Loading required package: zoo  
##  
## Attaching package: 'zoo'  
##  
## The following objects are masked from 'package:base':  
##  
##   as.Date, as.Date.numeric
```

```
Bank <- read_csv("BankFailures.csv") %>% as_tibble()
```

```
## Rows: 48 Columns: 6  
## -- Column specification -----  
## Delimiter: ","  
## chr (1): MONTH  
## dbl (5): YEAR, FAILURES, CPI, TOTAL.ASSETS, ADJTA  
##  
## i Use 'spec()' to retrieve the full column specification for this data.  
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
summary(Bank)
```

```
##      YEAR      MONTH      FAILURES      CPI  
## Min.   :2014   Length:48   Min.   :0.0000   Min.   :233.7  
## 1st Qu.:2015   Class :character 1st Qu.:0.0000   1st Qu.:237.1  
## Median :2016   Mode  :character  Median :1.0000   Median :238.3  
## Mean   :2016                      Mean   :0.8125   Mean   :239.7  
## 3rd Qu.:2016                      3rd Qu.:1.0000   3rd Qu.:242.0  
## Max.   :2017                      Max.   :3.0000   Max.   :246.8  
## TOTAL.ASSETS      ADJTA  
## Min.   :14121   Min.   :14121  
## 1st Qu.:15181   1st Qu.:15000  
## Median :15651   Median :15434  
## Mean   :15599   Mean   :15217  
## 3rd Qu.:16105   3rd Qu.:15554  
## Max.   :16788   Max.   :15930
```

```
Bank
```

```
## # A tibble: 48 x 6  
##   YEAR MONTH FAILURES  CPI TOTAL.ASSETS ADJTA  
##   <dbl> <chr>    <dbl> <dbl>    <dbl> <dbl>  
## 1  2014 Jan         3  234.    14121  14121
```

```
## 2 2014 Feb      2 235.      14257 14202.
## 3 2014 Mar      0 236.      14340. 14194.
## 4 2014 Apr      1 237       14393 14205.
## 5 2014 May      3 238.      14518. 14274.
## 6 2014 Jun      3 238.      14652. 14381.
## 7 2014 Jul      2 238.      14776. 14509.
## 8 2014 Aug      0 238.      14853. 14609.
## 9 2014 Sep      0 238       14933. 14676.
## 10 2014 Oct     2 237.      14975 14754.
## # i 38 more rows
```

```
Banks <- dummy_cols(Bank, select_columns = "MONTH", remove_first_dummy = TRUE)
Banks
```

```
## # A tibble: 48 x 17
##   YEAR MONTH FAILURES   CPI TOTAL.ASSETS ADJTA MONTH_Aug MONTH_Dec MONTH_Feb
##   <dbl> <chr>    <dbl> <dbl>      <dbl> <dbl>    <int>    <int>    <int>
## 1 2014 Jan      3 234.      14121 14121         0         0         0
## 2 2014 Feb      2 235.      14257 14202.         0         0         1
## 3 2014 Mar      0 236.      14340. 14194.         0         0         0
## 4 2014 Apr      1 237       14393 14205.         0         0         0
## 5 2014 May      3 238.      14518. 14274.         0         0         0
## 6 2014 Jun      3 238.      14652. 14381.         0         0         0
## 7 2014 Jul      2 238.      14776. 14509.         0         0         0
## 8 2014 Aug      0 238.      14853. 14609.         1         0         0
## 9 2014 Sep      0 238       14933. 14676.         0         0         0
## 10 2014 Oct     2 237.      14975 14754.         0         0         0
## # i 38 more rows
## # i 8 more variables: MONTH_Jan <int>, MONTH_Jul <int>, MONTH_Jun <int>,
## #   MONTH_Mar <int>, MONTH_May <int>, MONTH_Nov <int>, MONTH_Oct <int>,
## #   MONTH_Sep <int>
```

```
model1 <- glm(FAILURES ~ ADJTA + MONTH_Jan + MONTH_Feb + MONTH_Mar + MONTH_May + MONTH_Jun + MONTH_Jul +
summary(model1)
```

```
##
## Call:
## glm(formula = FAILURES ~ ADJTA + MONTH_Jan + MONTH_Feb + MONTH_Mar +
##   MONTH_May + MONTH_Jun + MONTH_Jul + MONTH_Aug + MONTH_Sep +
##   MONTH_Oct + MONTH_Nov + MONTH_Dec, family = "poisson", data = Banks)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) 11.4462863  4.4985580   2.544  0.01095 *
## ADJTA        -0.0007837  0.0003006  -2.607  0.00914 **
## MONTH_Jan     0.8216907  0.6902573   1.190  0.23388
## MONTH_Feb     0.3033143  0.7637682   0.397  0.69127
## MONTH_Mar    -0.4039310  0.9128715  -0.442  0.65814
## MONTH_May     0.8536216  0.6901173   1.237  0.21612
## MONTH_Jun     0.0250654  0.8168081   0.031  0.97552
## MONTH_Jul     0.1056980  0.8183228   0.129  0.89723
## MONTH_Aug    -0.9130919  1.1578700  -0.789  0.43035
## MONTH_Sep    -0.9139774  1.1582501  -0.789  0.43005
```

```
## MONTH_Oct    0.7509888  0.7392769   1.016  0.30970
## MONTH_Nov    -0.7550130  1.1653077  -0.648  0.51704
## MONTH_Dec    -0.0461541  0.9274979  -0.050  0.96031
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
## Null deviance: 58.152  on 47  degrees of freedom
## Residual deviance: 36.525  on 35  degrees of freedom
## AIC: 120.41
##
## Number of Fisher Scoring iterations: 6
```

```
model1
```

```
##
## Call:  glm(formula = FAILURES ~ ADJTA + MONTH_Jan + MONTH_Feb + MONTH_Mar +
##      MONTH_May + MONTH_Jun + MONTH_Jul + MONTH_Aug + MONTH_Sep +
##      MONTH_Oct + MONTH_Nov + MONTH_Dec, family = "poisson", data = Banks)
##
## Coefficients:
## (Intercept)      ADJTA    MONTH_Jan    MONTH_Feb    MONTH_Mar    MONTH_May
## 11.4462863   -0.0007837    0.8216907    0.3033143   -0.4039310    0.8536216
## MONTH_Jun    MONTH_Jul    MONTH_Aug    MONTH_Sep    MONTH_Oct    MONTH_Nov
##  0.0250654    0.1056980   -0.9130919   -0.9139774    0.7509888   -0.7550130
## MONTH_Dec
## -0.0461541
##
## Degrees of Freedom: 47 Total (i.e. Null);  35 Residual
## Null Deviance:      58.15
## Residual Deviance: 36.52    AIC: 120.4
```

```
round(exp(coef(model1)), 4)
```

```
## (Intercept)      ADJTA    MONTH_Jan    MONTH_Feb    MONTH_Mar    MONTH_May
## 93553.2745      0.9992      2.2743      1.3543      0.6677      2.3481
## MONTH_Jun    MONTH_Jul    MONTH_Aug    MONTH_Sep    MONTH_Oct    MONTH_Nov
##  1.0254      1.1115      0.4013      0.4009      2.1191      0.4700
## MONTH_Dec
##  0.9549
```

```
round(exp(confint(model1, level = 0.95)), 4)
```

```
## Waiting for profiling to be done...
```

```
##           2.5 %           97.5 %
## (Intercept) 11.9038 6.265782e+08
## ADJTA       0.9986 9.998000e-01
## MONTH_Jan   0.6320 1.055630e+01
## MONTH_Feb   0.2985 6.876300e+00
## MONTH_Mar   0.0879 4.029900e+00
```

```
## MONTH_May      0.6527 1.089660e+01
## MONTH_Jun      0.1897 5.543400e+00
## MONTH_Jul      0.2051 6.024700e+00
## MONTH_Aug      0.0198 3.156500e+00
## MONTH_Sep      0.0198 3.156600e+00
## MONTH_Oct      0.5113 1.048380e+01
## MONTH_Nov      0.0230 3.760300e+00
## MONTH_Dec      0.1232 5.927100e+00
```

```
(0.9986-1) * 100
```

```
## [1] -0.14
```

```
(0.9998-1) * 100
```

```
## [1] -0.02
```

```
model2 <- glm(FAILURES ~ ADJTA, family = "poisson", data = Banks)
summary(model2)
```

```
##
## Call:
## glm(formula = FAILURES ~ ADJTA, family = "poisson", data = Banks)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) 13.7047821  4.3189638   3.173  0.00151 **
## ADJTA        -0.0009216  0.0002885  -3.195  0.00140 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 58.152  on 47  degrees of freedom
## Residual deviance: 48.565  on 46  degrees of freedom
## AIC: 110.45
##
## Number of Fisher Scoring iterations: 5
```

```
model2
```

```
##
## Call:  glm(formula = FAILURES ~ ADJTA, family = "poisson", data = Banks)
##
## Coefficients:
## (Intercept)      ADJTA
## 13.7047821    -0.0009216
##
## Degrees of Freedom: 47 Total (i.e. Null);  46 Residual
## Null Deviance:      58.15
## Residual Deviance: 48.57    AIC: 110.5
```

```
anova(model2, model1, test = "Chisq")
```

```
## Analysis of Deviance Table
```

```
##
```

```
## Model 1: FAILURES ~ ADJTA
```

```
## Model 2: FAILURES ~ ADJTA + MONTH_Jan + MONTH_Feb + MONTH_Mar + MONTH_May +
```

```
##   MONTH_Jun + MONTH_Jul + MONTH_Aug + MONTH_Sep + MONTH_Oct +
```

```
##   MONTH_Nov + MONTH_Dec
```

```
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
```

```
## 1         46      48.565
```

```
## 2         35      36.525 11      12.04  0.3606
```